PUBLIC-PROJECT MECHANISM DESIGN • CERTIFICATE-FIRST RESEARCH

When the “fair” funding rule isn’t

An exact map of how small groups can break the textbook rule for funding shared projects — and by exactly how much.

Mohit Mendiratta · Independent research · SSRN Abstract ID 7293498

github.com/jammysunshine/when-fair-funding-isnt

THE PROBLEM, IN PLAIN TERMS

How do you fairly split the bill for something everyone benefits from?

- Picture an HOA board deciding whether to fund a new roof, a neighborhood group crowdfunding a footbridge, or a DAO voting to spend from a shared treasury.
- The standard textbook rule (the “pivotal mechanism”) says: build the project if pledges cover the cost, and charge each person only the smallest amount that was actually necessary for their vote to matter.
- This rule is provably honest for individuals — nobody benefits from lying about what they value the project at, acting alone.
- This paper asks the next question nobody had answered with an exact formula: what happens when a small group coordinates their pledges together?

THE HEADLINE FINDING

Groups can game the textbook rule — and this paper gives the exact formula for when

- The “fair” critical-payment rule is not fair against coordinated groups: in the tested cases, coalitions of just two or three people broke it in the large majority of configurations.
- This paper proves — not just tests — exactly when that happens: whenever the project's cost is low enough that any one member's pledge is dispensable, the whole group can jointly game the price to zero.
- A simple, checkable formula tells a designer, before launch, whether their specific group size, pledge cap, and project cost are exposed — and exactly how much a colluding group stands to gain.
- Every claim is backed by an executable, independently re-checked proof — not a simulation and not an opinion.

A CONCRETE EXAMPLE

Three neighbors, a $3,000 footbridge

Three neighbors truthfully value the bridge at $0, $1,000 and $1,000. Total pledges ($2,000) don’t cover the $3,000 cost, so honestly, nothing gets built and nobody pays anything.

But if the two $1,000 neighbors quietly agree to both claim $2,000 of value, the pledges now clear the threshold — the bridge gets built. Under the textbook payment rule, each of them is then charged $0, because each one's inflated claim alone was already “enough” once the other's inflated claim is counted.

- The two colluders get a $2,000 bridge for free.
- The honest third neighbor unknowingly ends up co-funding a project they never agreed the price was right for.
- This is not a bug in one bad rule — it is the standard, textbook “pivotal mechanism” taught in every mechanism-design course.

Exhaustive, machine-checked proof — not a sampled experiment

Exact enumeration. Every possible allocation rule in the studied finite class is checked — all of them, not a sample — against six formal fairness and honesty properties.

A proven theorem. A closed-form rule (not just a table of results) predicts exactly which allocation rules survive, for any group size and any pledge cap, and is confirmed against a larger untouched test domain.

Independent re-verification. A second, separately written program re-derives every result from the raw data without reusing any of the first program's code — catching the kind of error a single implementation could hide.

An exact formula for group manipulation. A closed-form expression for exactly how much a coalition of a given size can gain, derived by mathematical proof (convexity), not search — and it matches 145 independently brute-force-checked cases exactly.

A designer's checklist. The paper ends with a plain checklist: plug in your group size, pledge cap, and cost, and know before launch whether your specific setup is exposed.

SCOPE — STATED PLAINLY

What this paper is, and deliberately is not

IS

- An exact, fully-enumerated finite theorem
- A reproducible, independently re-checked certificate
- A precise formula for coalition risk on this specific rule
- A practical pre-launch checklist for real designers

IS NOT

- A new universal impossibility theorem
- A claim about continuous (non-integer) values
- A repaired, collusion-proof mechanism
- Evidence about any specific deployed platform

WHY IT MATTERS

Real systems already use this exact rule

- Participatory-budgeting platforms deciding which community proposal to fund.
- HOA and condo boards voting on a shared capital assessment.
- Crowdfunded neighborhood infrastructure — playgrounds, footbridges, shared solar installs.
- DAO treasuries and quadratic-funding platforms that have piloted pivotal-style payment rules.
- Anywhere a small, well-acquainted group decides together whether to build something and splits the bill by “what you were pivotal for” — this is the exact rule in question, and the paper's formula tells you if your setup is exposed.

Methodology and rigor

Certificate-first discipline. Every accepted or rejected rule ships with a machine-checkable witness; nothing is asserted from sampled performance.

Independent replay. A second implementation (public_project_independent.py) reconstructs every table and payment from serialized JSON without importing the primary verifier — zero mismatches across 145+ cross-checked rows.

SMT cross-checks. For the neural-network/ReLU audit lane, an external exact-real Z3 solver independently confirms all reported extrema — 18 strict-bound counterexample queries, all unsat.

Preregistration. The finite-lattice theorem and its confirmation domain were committed before the larger n=3, m=4 run; the coalition and false-name supplements are explicitly labeled post-hoc.

Full reproducibility. Every script, config, and SHA-256 digest is published; the entire study reruns from a single command list.

READ THE FULL PAPER

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papers.ssrn.com/sol3/papers.cfm?abstract_id=7293498

Full code, data, and independent verification scripts:

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